



First Semester 2018/2019
Lecture time:

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11
Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20	Q21	Q22
Q23	Q24	Q25								

$$d) 3xy - 2\sqrt{y^3} - 2\sqrt{x} = c$$
$$d) xe^{-t}(A + Bx) + C + Dx$$
d) $\frac{e^3}{x-3}$ d) $e^x(x-1)$

5. $\mathcal{L}^{-1}\left\{\frac{s-1}{s(2s+4)}\right\} =$

- a) $\frac{3}{4}e^{-2t} - \frac{1}{2}$ b) $\frac{3}{4}e^{2t} - \frac{1}{4}$ c) $\frac{3}{4}e^{-2t} + \frac{1}{4}$ d) none

6. If $f(t) = \mathcal{L}^{-1}\{\ln(s+2)^2\}$, then $f(2) =$

- a) e^{-4} b) $-e^{-4}$ c) $-e^4$ d) none

7. The solution of the IVP $xy' = y+1, y(1) = 2$, is $y =$

- a) $3x+1$ b) $x-1$ c) $3x-1$ d) $x+1$

8. Suppose that $y' = (x+y+1)^2$, then the general solution is $y =$

- a) $x + \tan(x+c)$ b) $x-1 + \tan^{-1}(x+c)$ c) $x-1 + \tan(x+c)$ d) none

9. A function $M(x,y)$ that makes $M(x,y)dx + (xe^y + 2xy + \frac{1}{x})dy = 0$

to be exact equals

- a) $ye^y + y^2 - \frac{y}{x^2} + h(y)$ b) $ye^y + y^2 - \frac{y}{x^2} + h(x)$
c) $xye^y + e^y + 2y - \frac{1}{x} + h(y)$ d) $xye^y + e^y + 2y - \frac{1}{x} + h(x)$

10. If $\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 4 & 1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$, where $\lambda = 3,3$, such that $X_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{3t}$, then $X_2 =$

- a) $\begin{pmatrix} 1 \\ -1 \end{pmatrix} t e^{3t} + \begin{pmatrix} -1 \\ 1 \end{pmatrix} e^{3t}$ b) $\begin{pmatrix} 1 \\ -1 \end{pmatrix} t e^{3t} + \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ c) $\begin{pmatrix} 1 \\ -1 \end{pmatrix} t e^{3t} + \begin{pmatrix} -1 \\ 0 \end{pmatrix} e^{3t}$ d) none

11. The orthogonal trajectories for $y = \ln(\tan x + c)$ are $e^{-y} =$

- a) $\frac{x}{2} + \frac{1}{4} \sin 2x + c$ b) $\frac{x}{2} - \frac{1}{4} \sin 2x + c$ c) $\tan x + c$ d) $-\tan x + c$

12- If $y = c_1 e^x + c_2 e^{-2x} + e^{-x} + 1$ is the G.S. of $y'' + Ay' + By = g(x)$, then $g(x) =$

- a) $-2e^{-x} - 2$ b) $-2e^{-x} + 2$ c) $2e^{-x} + 2$ d) -2

13- If the G.S. of $x^3 y'' + ax^2 y' + bxy' + cy = 0$ is $y = c_1 x^2 + c_2 x^2 \ln x + c_3 x^{-1}$, then $a =$

- a) 3 b) -2 c) 0 d) 2

14- If $X' = \begin{pmatrix} 0 & 4 \\ -1 & 0 \end{pmatrix} X$, such that the eigenvalue is $\lambda = \pm 2i$, then $X_1 =$

- a) $\begin{pmatrix} 1 \\ 2 \end{pmatrix} \cos 2t - \begin{pmatrix} 4 \\ 0 \end{pmatrix} \sin 2t$ b) $\begin{pmatrix} 0 \\ 4 \end{pmatrix} \cos 2t - \begin{pmatrix} 0 \\ 2 \end{pmatrix} \sin 2t$
c) $\begin{pmatrix} 4 \\ 0 \end{pmatrix} \cos 2t - \begin{pmatrix} 2 \\ 0 \end{pmatrix} \sin 2t$ d) $\begin{pmatrix} 4 \\ 0 \end{pmatrix} \cos 2t - \begin{pmatrix} 0 \\ 2 \end{pmatrix} \sin 2t$

15- A homogenous linear differential equation of order 2 one of its solution is $y_1 = x$, and the Wronskian's $W(y_1, y_2) = x$, then $y_2 =$

- a) $\frac{1}{2}x^3 + cx$ b) $x \ln x + cx$ c) $x \ln x + c$ d) $\frac{1}{2}x^3 + c$

16- The $\mathcal{L}\left\{\int_0^t \sin(\tau) d\tau\right\} =$

- a) $\frac{1}{s(s^2 - 1)}$ b) $\frac{1}{(s^2 - 1)}$ c) $\frac{1}{s(s^2 + 1)}$ d) $\frac{1}{(s^2 + 1)}$

17- The inverse Laplace of $\left\{\frac{s}{2s^2 + 4s + 6}\right\}$ equals

- a) $\frac{1}{2}e^{-t} \cos 2t$ b) $\frac{1}{2}e^{-t} \cos \sqrt{2}t$ c) $\frac{1}{2\sqrt{2}}e^{-t} \cos \sqrt{2}t$ d) none

18- If $y'' + y = \sinh t$, $y(0) = 0$, $y'(0) = 1$, such that $\mathcal{L}\{y(t)\} = Y(s)$, then $Y(2) =$

- a) $\frac{1}{3}$ b) $\frac{4}{15}$ c) $\frac{1}{15}$ d) $\frac{3}{15}$

- 19- If $y = \sum_{n=0}^{\infty} a_n x^n$ is the general solution of $2y'' + 4y' = 0$, then $a_2 =$
 a) a_1 b) $-a_1$ c) $-2a_0$ d) $-a_0$
- 20- If $Y(s) = \mathcal{L}\{y(t)\} = \ln(s+1)$, and $\mathcal{L}\{ty'(t)\} = Y_1(s)$, $y(0) = \ln 2$, then $Y_1(1) =$
 a) $\frac{1}{2}$ b) $0.5 - \ln 2$ c) $-\frac{1}{2}$ d) $-0.5 - \ln 2$
- 21- If $f(t) = \mathcal{L}^{-1}\left\{\frac{2}{s(s+1)^2}\right\}$, then $f(1) =$
 a) $2 - 4e^{-1}$ b) $2 + 4e^{-1}$ c) $1 - 4e^{-1}$ d) $1 + 4e^{-1}$
- 22- The integrating factor of $\sin(x)y' + 2\cos(x)y + \sin(x)y = e^{-x}$ is
 a) $\sin^2 x + x^2$ b) $\sin^2 x$ c) $e^x \sin^2 x$ d) $e^x \sin x$
- 23- If $F(s) = \mathcal{L}\{e^{-3t} f(t)\} = \frac{3}{(s+3)^4}$, then $f(t) =$
 a) t^3 b) $\frac{1}{2}t^3$ c) $\frac{1}{8}t^4$ d) $\frac{1}{6}t^3$
- 24- If the general solution of a D.E., using the power series solution, is
 $y = a_0 \sum_{n=0}^{\infty} (-1)^n (3x)^n + a_1 \sum_{n=0}^{\infty} x^n$, then this solution can be represented as $y =$
 a) $a_0 e^{-3x} + \frac{a_1}{1-x}$ b) $a_0 e^{-3x} + a_1 e^x$ c) $\frac{a_0}{1+3x} + \frac{a_1}{1-x}$ d) none
- 25- To solve the nonlinear (Bernoulli) D.E. $xy' + y = x^3 y^2$, we convert it to
 L.D.E., then the integrating factor of the resulting Linear D.E. is $\mu =$
 a) x^{-1} b) x c) x^3 d) none

BEST WISHES

السئلة سنوات سابقة

المادة:

$$Q1) y' = \frac{\sqrt{x} - y}{x - \sqrt{y}}$$

$$\frac{dy}{dx} = \frac{\sqrt{x} - y}{x - \sqrt{y}}$$

$$\underbrace{(\sqrt{x} - y) dx}_M - \underbrace{(x - \sqrt{y}) dy}_N = 0$$

$$\frac{\partial M}{\partial y} = -1$$

$$\frac{\partial N}{\partial x} = -1$$

$$\int \sqrt{x} - y \, dx = \frac{2}{3} x^{\frac{3}{2}} - yx + f(y) \quad \dots \text{a.o.}$$

$$\frac{\partial F}{\partial y} = \cancel{-x} + f'(y) = \cancel{-x} + \sqrt{y}$$

$$g'(y) = \sqrt{y}$$

$$g(y) = \int \sqrt{y} \, dy = \frac{2}{3} y^{\frac{3}{2}}$$

$$3 \left(\frac{2}{3} \sqrt{x^3} - yx + \frac{2}{3} \sqrt{y^3} = C \right)$$

$$-1 \cdot (2\sqrt{x^3} - yx + 2\sqrt{y^3} = C)$$

$$3xy - 2\sqrt{y^3} - 2\sqrt{x^3} = C \quad \dots \quad (C)$$

الأسئلة لسنوات سابقة

المادة:

Q2) $y''' + y'' - 4y' - 4y = 2x e^{-x} + x$

$$\lambda^3 + \lambda^2 - 4\lambda - 4 = 0$$

$$(\lambda - 2)(\lambda^2 + 3\lambda + 2) = 0$$

$$(\lambda - 2)(\lambda + 1)(\lambda + 2) = 0$$

$$\lambda = -1, 2, -2$$

$$y' = \frac{1}{e^{-x}} \frac{1}{e^{2x}} \frac{1}{e^{-2x}}$$

$$\lambda = -1, -1, 0, 0$$

$$\lambda = -1, -1, -1, 2, -2, 0, 0$$

$$e^x x e^{-x} x^2 e^x x^2 e^{-x} 1 x$$

$$2 \begin{vmatrix} 1 & 1 & -4 & -4 \\ & 2 & 6 & 4 \\ & & 3 & 2 \\ & & & 0 \end{vmatrix}$$

$$y_p = Ax e^{-x} + Bx^2 e^{-x} + C + Dx$$

$$y_p = x e^{-x} (A + Bx) + C + Dx$$

(d)

Q3) $\int e^{-3(t-1)} dt = \int e^{-3t} e^3 dt$

$$= e^3 \int e^{-3t} dt = \frac{e^3}{-3} e^{-3t} = -\frac{e^3}{3} e^{-3t}$$

(b)

Q4) $y_p = y_1 u_1 + y_2 u_2$

$$x^2 y'' - 2y = 3x e^x$$

$$y_1 = x^{-1} \quad y_2 = x^2$$

$$y'' - \frac{2}{x^2} y = \frac{3}{x} e^x$$

$$u_1 = \int \frac{3e^x (-x^2)}{3x} dx = -\int x e^x dx$$

$$w_1(x) = \begin{vmatrix} \frac{1}{x} & x^2 \\ -\frac{1}{x^2} & 2x \end{vmatrix} = 2 - (-1) = 3$$

$$u_1 = -x e^x + e^x$$

$$u_1 = e^x (1 - x)$$

(a)

$$\begin{vmatrix} x & e^x \\ 1 & e^x \\ 0 & e^x \end{vmatrix}$$

Don't miss the change ... Be part of it

اسئلة سنوات سابقة

المادة:



$$Q5) \mathcal{L}^{-1} \left\{ \frac{s-1}{s(2s+4)} \right\}$$

$$\mathcal{L}^{-1} \left\{ \frac{s-1}{2s^2+4s} \right\}$$

$$= \frac{A}{2s} + \frac{B}{s+2}$$

$$s-1 = A(s+2) + 2Bs$$

$$s=0 \quad -1 = A \quad A = -\frac{1}{2}$$

$$s=-2 \quad -3 = -4B \quad B = \frac{3}{4}$$

$$\mathcal{L}^{-1} \left\{ \frac{-\frac{1}{2}}{2s} + \frac{\frac{3}{4}}{s+2} \right\} = \mathcal{L}^{-1} \left\{ \frac{-1}{4s} + \frac{\frac{3}{4}}{s+2} \right\}$$

$$= \frac{3}{4} e^{-2t} - \frac{1}{4} \quad \dots \textcircled{b}$$

$$Q6) f(t) = \mathcal{L}^{-1} \{ \ln(s+2) \}$$

$$f(2) = ?$$

$$F(s) = \ln(s+2)^2$$

$$F'(s) = \frac{2(s+2)}{(s+2)^2} = \frac{2}{s+2}$$

$$\mathcal{L}^{-1} \left\{ \frac{2}{s+2} \right\} = 2e^{-2t}$$

$$f(t) = -\frac{1}{t} 2e^{-2t}$$

$$f(2) = -\frac{1}{2} 2e^{-4} = -e^{-4} \quad \dots \textcircled{b}$$



أسئلة سنوات سابقة

المادة:

Q 7) $xy' = y + 1$, $y(1) = 2$ $y = ?$

$$x \frac{dy}{dx} = y + 1$$

$$\int \frac{dy}{y+1} = \int \frac{dx}{x}$$

$$\ln(y+1) = \ln x + C$$

$$y+1 = e^{\ln x + C} = Cx$$

$$C = 3$$

$$y = 3x - 1 \quad \text{--- (C)}$$

Q 8) $y' = (x+y+1)^2$

$$z = x + y + 1$$

$$z' = 1 + y'$$

$$y' = z' - 1$$

$$z' - 1 = z^2$$

$$z' = z^2 + 1$$

$$\frac{dz}{dx} = z^2 + 1$$

$$\int \frac{dz}{z^2 + 1} = \int \frac{1}{dx}$$

$$\tan^{-1} z = x + C$$

$$z = \tan(x + C)$$

$$x + y + 1 = \tan(x + C)$$

$$y = -x - 1 + \tan(x + C) \quad \text{--- (d)}$$



امثلة سنوات سابقة

المادة:

$$\textcircled{9} \quad M(x, y) dx + (x e^{xy} + 2xy + \frac{1}{x}) dy = 0$$

$$\int x e^{xy} + 2xy + \frac{1}{x} dy$$

$$e^{xy} + xy^2 + \frac{1}{x} y + g(x)$$

$$\frac{df}{dx} = y e^{xy} + y^2 - \frac{y}{x^2} + g'(x)$$

$$M(x, y) = y e^{xy} + y^2 - \frac{y}{x^2} + h(x) \quad \text{---} \quad \textcircled{b}$$

$$\textcircled{10} \quad \begin{vmatrix} x' \\ y' \end{vmatrix} = \begin{vmatrix} 4 & 1 \\ -1 & 2 \end{vmatrix} \begin{vmatrix} x \\ y \end{vmatrix} \quad \lambda = 3, 3 \quad x_1 = \begin{vmatrix} 1 \\ -1 \end{vmatrix} e^{3t}$$
$$x_2 = ?$$

$$\begin{vmatrix} 1 & 1 \\ -1 & -1 \end{vmatrix} \begin{vmatrix} b_1 \\ b_2 \end{vmatrix} = \begin{vmatrix} 1 \\ -1 \end{vmatrix}$$

$$b_1 + b_2 = 1$$

$$b_2 = 0$$

$$b_1 = 1$$

$$x_2 = \begin{vmatrix} 1 \\ -1 \end{vmatrix} t e^{3t} + \begin{vmatrix} 1 \\ 0 \end{vmatrix} e^{2t} \quad \text{---} \quad \textcircled{d}$$



أسئلة سنوات سابقة

العادة:

Q 11) $y = \ln(\tan x + c)$, $e^{-y} = ?$

$$y' = \frac{\sec^2 x}{\tan x + c}$$

$$e^y = \tan x + c$$

$$c = e^y - \tan x$$

$$y' = \frac{\sec^2 x}{\tan x + e^y - \tan x} = \frac{\sec^2 x}{e^y}$$

$$\frac{dy}{dx} = \frac{\sec^2 x}{e^y}$$

$$\frac{dy}{dx} = -\frac{e^y}{\sec^2 x}$$

$$\int -e^y dy = \int \cos^2 x dx$$

$$e^{-y} = \frac{x}{2} + \frac{1}{4} \sin 2x + c \quad \text{--- (a)}$$

Q 12) $y = \underbrace{c_1 e^x + c_2 e^{-x}}_{y_c} + \underbrace{e^{-x} + 1}_{y_p}$

$$(\lambda - 1)(\lambda + 2) = 0$$

$$\lambda^2 + \lambda + 2\lambda - 2 = 0$$

$$\lambda^2 - \lambda - 2 = 0$$

$$y'' - y' - 2y = R(x)$$

$$y_p = e^{-x} + 1$$

$$y_p' = -e^{-x}$$

$$y_p'' = e^{-x}$$

$$e^{-x} + (-e^{-x}) - 2(e^{-x} + 1) = R(x)$$

$$R(x) = -2 \quad \text{--- (d)}$$

Don't miss the change ... Be part of it

امثلة سنوات سابقة

المادة:



Q14) $x' = \begin{vmatrix} 0 & 4 \\ -1 & 0 \end{vmatrix} x$ $\lambda = \pm 2i$ $x_1 = ?!$

$$\begin{vmatrix} -2i & 4 \\ -1 & -2i \end{vmatrix} \begin{vmatrix} a_1 \\ a_2 \end{vmatrix} = \begin{vmatrix} 0 \\ 0 \end{vmatrix}$$

$$-2i a_1 + 4 a_2 = 0$$

$$a_1 = \frac{4 a_2}{2i}$$

$$a_2 = 1 + 2i$$

$$a_1 = 4$$

$$\begin{vmatrix} 4 \\ -2i \end{vmatrix}$$

$$x_1 = \begin{vmatrix} 4 \\ 0 \end{vmatrix} \cos 2t - \begin{vmatrix} 0 \\ 2 \end{vmatrix} \sin 2t \quad \text{--- (d)}$$

Q15) $y_1 = x$ $w(y_1, y_2) = x$ $y_2 = ?!$

$$\begin{vmatrix} x & y_2 \\ 1 & y_2' \end{vmatrix} = x$$

$$x y_2' - y_2 = x$$

$$y_2' - \frac{1}{x} y_2 = 1$$

$$M = e^{\int -\frac{1}{x} dx} = x^{-1}$$

$$y = x \int \frac{1}{x} dx = x (\ln x + c) = x \ln x + cx$$

Q16) $\mathcal{L} \left\{ \int_0^t \sin \tau d\tau \right\} = \frac{f(s)}{s}$ --- (b)

$$F(s) = \mathcal{L} \{ \sin t \} = \frac{1}{s^2 + 1} = \frac{1}{s(s^2 + 1)} \quad \text{--- (c)}$$

Don't miss the change ... Be part of it

أسئلة سنوات سابقة المادة:



$$Q17) \mathcal{L}^{-1} \left\{ \frac{s}{2s^2 + 4s + 6} \right\}$$

$$\frac{1}{2} \mathcal{L}^{-1} \left\{ \frac{s}{s^2 + 2s + 3} \right\}$$

$$\frac{1}{2} \mathcal{L}^{-1} \left\{ \frac{s}{s^2 + 2s + 1 + 2} \right\}$$

$$\frac{1}{2} \mathcal{L}^{-1} \left\{ \frac{s}{(s+1)^2 + 2} \right\}$$

$$\frac{1}{2} e^{-t} \cos \sqrt{2} t \quad \text{--- (b)}$$

$$s^2 + 2s + 3s = 0$$

$$s^2 + 2s + 1 - 2$$

$$Q18) s^2 y(s) - s y(0) - y'(0) + y(s) = \mathcal{L} \{ \sinh x \}$$

$$s^2 y(s) - 1 + y(s) = \frac{1}{s^2 + 1}$$

$$y(s) (s^2 + 1) = \frac{1}{s^2 + 1} \times 1$$

$$y(s) = \frac{s^2}{(s^2 + 1)(s^2 + 1)}$$

$$y(s) = \frac{4}{15} \quad \text{--- (b)}$$



أسئلة سنوات سابقة

المادة:

Q 19) $y = \sum_{n=0}^{\infty} z_n x^n$

$$2y'' + 4y' = 0$$

$$z_2 = ?!$$

$$y' = \sum_{n=1}^{\infty} n z_n x^{n-1}$$

$$y'' = \sum_{n=2}^{\infty} z_n n(n-1) x^{n-2}$$

$$\sum_{n=2}^{\infty} 2n(n-1) z_n x^{n-2} + 4 \sum_{n=1}^{\infty} n z_n x^{n-1}$$

$$m = n-2$$

$$m = n-1$$

$$n = m+2$$

$$n = m+1$$

$$n=2 \rightarrow m=0$$

$$n=1 \rightarrow m=0$$

$$\sum_{m=0}^{\infty} 2(m+2)(m+1) z_{m+2} x^m + \sum_{m=0}^{\infty} 4(m+1) z_{m+1} x^m$$

$$\sum_{m=0}^{\infty} (2(m+2)(m+1) z_{m+2} + 4(m+1) z_{m+1}) x^m$$

$$2(m+2)(m+1) z_{m+2} = -4(m+1) z_{m+1}$$

$$z_{m+2} = \frac{-2(m+1) z_{m+1}}{(m+2)(m+1)}$$

$$z_{m+2} = \frac{-2 z_{m+1}}{m+2}$$

$$z_2 = \frac{-2 z_1}{2}$$

$$z_2 = -z_1$$

--- (b)

Q 20) $\mathcal{L}\{y(x)\} = \ln(s+1)$



أسئلة سنوات سابقة

المادة:

$$Q21) \mathcal{L}^{-1} \left\{ \frac{2}{s(s+1)^2} \right\}$$

$$f(t) = ?$$

$$\int_0^t 2\tau e^{-\tau} d\tau$$

$$-2\tau e^{-\tau} - 2e^{-\tau} \Big|_0^t$$

$$-2te^{-t} - 2e^{-t} + 2$$

$$f(t) = 2 - 4e^{-t} \quad \text{--- (a)}$$

$$\begin{array}{r} 2\tau \quad e^{-\tau} \\ 2 \quad \swarrow + -e^{-\tau} \\ 0 \quad \searrow e^{-\tau} \end{array}$$

$$Q22) \sin x \frac{dy}{dx} = -y(2\cos x + \sin x) + e^{-x}$$

$$-e^{-x} + \underbrace{y(2\cos x + \sin x)}_P dx + \underbrace{\sin x dy}_Q = 0$$

$$\frac{dP}{dy} = 2\cos x + \sin x \quad \frac{dQ}{dx} = \cos x$$

$$\frac{dP}{dy} - \frac{dQ}{dx} = 2\cos x + \sin x - \cos x$$
$$= \cos x + \sin x$$

$$\frac{\cos x + \sin x}{\sin x} = \cot x + 1$$

$$M = e^{\int \cot x + 1 dx} = e^{(\ln \sin x + x)}$$
$$= e^x \sin x \quad \text{--- (d)}$$



أسئلة سنوات سابقة

العادة:

$$Q22) \mathcal{L}\{e^{-3t} f(t)\} = \frac{3}{(s+3)^4}$$

$$e^{-3t} f(t) = \mathcal{L}^{-1}\left\{\frac{3}{(s+3)^4}\right\}$$

$$e^{-3t} f(t) = \frac{3 e^{-3t} t^3}{3!}$$

$$f(t) = \frac{1}{2} t^3 \quad \text{--- (b)}$$

$$Q24) \frac{z_0}{1+3x} + \frac{z_1}{1-x} \quad \text{--- (c)}$$

$$Q25) M = e^{\int \frac{1}{x} dx}$$

$$= e^{\ln x} = x \quad \text{--- (b)}$$